

2

Digital Logic

Practice Questions

Number System

- Q.1** 4-bit 2's complement representation of a decimal number is 1000. The number is
(A) +8 (B) 0
(C) -7 (D) -8
- Q.2** Decimal 43 in Hexadecimal and BCD number system is respectively
(A) B2, 01000011
(B) 2B, 01000011
(C) 2B, 0011 0100
(D) B2, 01000100
- Q.3** $X = 01110$ and $Y = 11001$ are two 5-bit binary numbers represented in two's complement format. The sum of X and Y represented in 2's complement format using 6 bits is
(A) 100111 (B) 001000
(C) 000111 (D) 101001
- Q.4** The two numbers represented in signed 2's complement form are
 $P = 11101101$ and $Q = 11100110$
If Q is subtracted from P , the value obtained in signed 2's complement form is
(A) 100000111 (B) 00000111
(C) 11111001 (D) 11111001
- Q.5** What is the gray code word for the binary number 101011?
(A) 110101 (B) 111110
(C) 101011 (D) 011111

- Q.6** Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions are _____.
- Q.7** If $(101)_b + (D2)_{16} = (467)_{10}$. Then the value of radix ' b ' is _____.
(A) 13 (B) 14
(C) 18 (D) 16
- Q.8** The number $(A72E)_{16}$ is equivalent to
(A) $(1010001100101110)_2$ and $(123456)_8$
(B) $(1010011100111111)_2$ and $(123456)_8$
(C) $(1010011100101110)_2$ and $(123456)_8$
(D) None of these

Boolean Algebra

- Q.9** The K-map for a Boolean function is shown below. The number of essential prime implicants for this function is

		AB			
		00	01	11	10
CD	00	1	1	0	1
	01	0	0	0	1
	11	1	0	0	0
	10	1	0	0	1



- (A) 4 (B) 5
(C) 6 (D) 8
- Q.10** The logical expression $y = A + \bar{A}B$ is equivalent to
(A) $y = AB$ (B) $y = \bar{A}B$
(C) $y = \bar{A} + B$ (D) $y = A + B$
- Q.11** The minimized form of the logical expression $(\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + ABC\bar{C})$ is
(A) $\bar{A}\bar{C} + B\bar{C} + \bar{A}B$
(B) $A\bar{C} + \bar{B}C + \bar{A}B$
(C) $\bar{A}C + \bar{B}C + \bar{A}B$
(D) $A\bar{C} + \bar{B}C + A\bar{B}$
- Q.12** The number of distinct Boolean expression of 4 variables is
(A) 16 (B) 256
(C) 1024 (D) 65536
- Q.13** The Boolean expression $Y = \bar{A}\bar{B}\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + ABC\bar{D}$ can be minimized to
(A) $Y = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C} + A\bar{C}D$
(B) $Y = \bar{A}\bar{B}\bar{C}D + BC\bar{D} + A\bar{B}\bar{C}D$
(C) $Y = \bar{A}BC\bar{D} + \bar{B}\bar{C}D + A\bar{B}\bar{C}D$
(D) $Y = \bar{A}BC\bar{D} + \bar{B}\bar{C}D + ABC\bar{D}$
- Q.14** The number of essential prime implicants in the function $f(a, b, c, d) = \Sigma(1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15)$ is _____.
- Q.15** The Boolean expression $(X + Y)(X + \bar{Y}) + \overline{(X\bar{Y})} + \bar{X}$ simplifies to
(A) X (B) Y
(C) XY (D) $X + Y$

- Q.16** Consider the Boolean function,
 $F(w, x, y, z) = wy + xy + \bar{w}xyz + \bar{w}\bar{x}y + xz + \bar{x}\bar{y}\bar{z}$

Which one of the following is the complete set of essential prime implicants?

- (A) $w, y, xz, \bar{x}\bar{z}$ (B) w, y, xz
(C) $y, \bar{x}\bar{y}\bar{z}$ (D) $y, xz, \bar{x}\bar{z}$
- Q.17** Find how many number of literals are there in given K-map by using all possible type K-map?

		CD	
AB			
		1	
		1	1
	1	1	1
			1
		1	1

- (A) 4 (B) 14
(C) 17 (D) 8
- Q.18** A function of Boolean variables X, Y and Z is expressed in terms of the minterms as $F(X, Y, Z) = \Sigma m(1, 2, 5, 6, 7)$
Which one of the product of sums given below is equal to the function $F(X, Y, Z)$?
(A) $(\bar{X} + \bar{Y} + \bar{Z})(\bar{X} + Y + Z)(X + \bar{Y} + \bar{Z})$
(B) $(X + Y + Z)(X + \bar{Y} + \bar{Z})(\bar{X} + Y + Z)$
(C) $(\bar{X} + \bar{Y} + Z)(\bar{X} + Y + \bar{Z})(X + \bar{Y} + \bar{Z})$
(D) $(X + Y + \bar{Z})(\bar{X} + Y + Z)(\bar{X} + \bar{Y} + \bar{Z})$

- Q.19** Consider the following expression :
 $f(A, B, C, D) = AD + ABCD + ACD + \bar{A}B + A\bar{C}D + \bar{A}\bar{B} + \bar{A}(\bar{B} + A)$
The minimized expression of $f(A, B, C, D)$ is equal to